Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Fundamental of Computer **Subject Code : CSE-101**

Time Allowed : 3 Hours **Full Marks : 70**

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory*

The figures in the margin indicate full marks.

All questions carry equal marks

2×7=14

1. Choose the correct alternative.

(i) The binary equivalent of decimal 13 is

(a) 1100 (b) 1011

(c) 1110 (d) 1001

(ii) A flip-flop is used for

(a) combinational logic (b) storing 1 bit of data.

(c) performing arithmetic operations. (d) multiplexing signals.

(iii) Which of the following is an example of system software?

(a) MS Word (b) Windows OS

(c) Chrome browser (d) Photoshop

(iv) Which memory is fastest among the following?

(a) Cache memory (b) Main memory

(c) Secondary memory (d) Virtual memory

(v) Which symbol is used in a flowchart to represent a decision?

(a) Rectangle (b) Oval

(c) Diamond (d) Parallelogram

(vi) A Real-Time Operating System (RTOS) is mainly used in

(a) banking systems. (b) scientific computations.

(c) embedded systems (d) general-purpose computers.

(vii) The main difference between synchronous and asynchronous sequential circuits is:

(a) Synchronous uses a clock signal. (b) Asynchronous uses a clock signal.

(c) Both uses a clock signal. (d) None of these

2. (a) Explain the concept of 1's and 2's complements. Perform subtraction of two binary numbers using 2's complement method.

   (b) Describe BCD, ASCII and Gray codes and their applications in digital systems. 7+7

3. (a) Explain the characteristics and types of flip-flops (SR, JK, D, T) and draw their truth tables.

   (b) Draw a block diagram of a combinational and sequential circuit and explain their working. 7+7

4. (a) Describe the different methods of data processing: single-user, multiprogramming, real-time, online, time-sharing and distributed processing.

   (b) Explain the evolution of computers and the characteristics of each generation. 7+7

5. (a) Discuss the types of operating systems: batch, multitasking, multiprocessing, real-time, distributed, network and mobile OS.

   (b) Explain Flynn's classification of computers with examples. 7+7

6. (a) Define an algorithm and explain its characteristics. Write an algorithm to find the factorial of a number.

   (b) Write an algorithm and flowchart to check whether a number is prime or not. 7+7

7. Write short notes *any four*: 3.5×4=14

   (a) Logic Gates and Boolean Algebra

   (b) Multiplexers and Demultiplexers

   (c) Classification of Computers and Their Applications

   (d) BIOS and UEFI

   (e) Algorithms and Their Characteristics